

Representation Theorem for finite Boolean algebras

We now define structure-preserving maps between Boolean algebras.

Definition 0.1. (Homomorphism)

Consider the Boolean algebras $\mathcal{B} := (B, \leq, \vee, \wedge, ^c, 0, 1)$ and $\mathcal{B}' := (B', \leq, \vee, \wedge, ^c, 0, 1)$, and a map h from B to B' .

h is a *homomorphism* from $\mathcal{B}$ to $\mathcal{B}'$ if and only if for all x, y in B ,

$$(i) \quad h(x \wedge y) = h(x) \wedge h(y) \text{ and}$$

$$(ii) \quad h(x^c) = (h(x))^c.$$

A homomorphism h is an *isomorphism*, if it is a bijection.

Note. We have used the same notation for the operations in $\mathcal{B}$ and $\mathcal{B}'$. The context should make the domain of the operations clear.

Exercise 0.1. For any homomorphism h from $\mathcal{B}$ to $\mathcal{B}'$, prove that

$$h(x \vee y) = h(x) \vee h(y),$$

$$h(0) = 0 \text{ and}$$

$$h(1) = 1.$$

Proposition 0.1. Let h be an onto map from B to B' , where $\mathcal{B} := (B, \leq, \vee, \wedge, ^c, 0, 1)$ and $\mathcal{B}' := (B', \leq, \vee, \wedge, ^c, 0, 1)$ are Boolean algebras.

h is an isomorphism from $\mathcal{B}$ to $\mathcal{B}'$ if and only if for all $x, y \in B$,

$$x \leq y \text{ if and only if } h(x) \leq h(y). \quad (*)$$

The proof of the proposition is fairly routine.

Exercise 0.2. Let $\mathcal{B}$ and $\mathcal{B}'$ be isomorphic Boolean algebras. If one is atomic, the other is also so; if one is atomless, the other is also so.

Theorem 0.2. (Representation Theorem 1)

Every finite Boolean algebra is isomorphic to the power set Boolean algebra of some set.

Proof. Let $\mathcal{B} := (B, \leq, \vee, \wedge, ^c, 0, 1)$ be a finite Boolean algebra. By earlier proposition, $\mathcal{B}$ is atomic.

Let A be the set of its atoms. (A is non empty...why?).

We claim that $\mathcal{B}$ is isomorphic to the power set Boolean algebra of A .

Define $h : B \longrightarrow \mathcal{P}(A)$ to map an element x of B to the set of atoms below x . Formally, for each $x \in B$,

$$h(x) := \{a \in A : a \leq x\}.$$

By Proposition 0.1, it is enough to show that

- (i) h is onto, and
- (ii) h satisfies the condition $(*)$ of the proposition.

(i) Observe that $h(0) = \emptyset$.

Now consider $(\emptyset \neq) S := \{a_1, a_2, \dots, a_n\} \subseteq A$, and let $x := a_1 \vee \dots \vee a_n$.

Claim: $h(x) = S$.

Proof of Claim: $S \subseteq h(x)$ follows from definition.

For proving $h(x) \subseteq S$, let $a \in h(x)$, i.e. $a \leq a_1 \vee \dots \vee a_n$.

We show that $a \leq a_i$, for some $i \in \{1, \dots, n\}$ (since a and a_i are both atoms, it would follow that $a = a_i$).

Argue by contradiction: if $a \not\leq a_i$ for all $i \in \{1, \dots, n\}$, then by previous Exercise, $a \leq a_i^c$ for all $i \in \{1, \dots, n\}$. This would give $a \leq a_1^c \wedge \dots \wedge a_n^c = (a_1 \vee \dots \vee a_n)^c$. But then $a = 0$, a contradiction to the fact that a is an atom. Thus $a \in S$.

(ii) For any $x, y \in B$, if $x \leq y$, then it is clear that $h(x) \subseteq h(y)$. Conversely, let $h(x) \subseteq h(y)$, $x, y \in B$. Suppose $x \not\leq y$. Then $x \wedge y^c \neq 0$, by earlier proposition. As $\mathcal{B}$ is atomic, there is an atom $b \in A$ below $x \wedge y^c$, i.e. $b \leq x \wedge y^c$. $b \leq x$ implies that $b \in h(x)$, while $b \leq y^c$ implies that $b \not\leq y$ (as $b \neq 0$) – which means that $b \notin h(y)$. So $h(x) \not\subseteq h(y)$. □

Corollary 0.3. The cardinality of B , $|B| = 2^{|A|}$. In other words, the cardinality of any finite Boolean algebra is a power of 2.

So is every Boolean algebra isomorphic to some power set Boolean algebra?

A little thought will tell us that it cannot be the case:

Exercise 0.2 would then entail that all Boolean algebras are atomic, as the power set Boolean algebras are so. Next, we present the representation theorem in the general case.